Total No. of Questions : 8]

SEAT No. :

PC-1685

[Total No. of Pages : 2

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T.E. (Civil)
Hydrology and Water Resources Engineering
(2019 Pattern) (Semester-I) (301001)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume Suitable data if necessary.***

Q1) a) Explain Q-GIS and its application in hydrology **[10]**

b) Explain Rational formula and its importance **[8]**

OR

Q2) a) Explain watershed delineation procedure using a topo sheet with neat sketches **[10]**

b) Explain flood routing in detail **[8]**

Q3) a) Explain how will you fix the capacity of reservoir using annual inflow and outflow **[10]**

b) What are reservoir losses and suggest method to control leakages from reservoir. **[7]**

OR

Q4) a) What are various investigations required for reservoir planning **[10]**

b) State measures to control reservoir sedimentation **[7]**

P.T.O.

Q5) a) Derive the formula to calculate discharge of a well in a confined aquifer and unconfined aquifer. **[10]**

b) What is water logging? Explain tile drain method and also state formula for spacing of tile drains. **[8]**

OR

Q6) a) Explain reclamation of saline lands **[10]**

b) State various types of tube wells and explain construction of slotted type tube well **[8]**

Q7) a) Explain Piped Distribution Network (PDN) and state its advantages **[10]**

b) Explain Hortons curve with neat sketch. **[7]**

OR

Q8) a) State principle Indian crops and explain their planning and agricultural practices **[2+4+4]**

b) Differentiate between surface irrigation and subsurface irrigation and explain drip irrigation in detail **[7]**